Exam Visual Analytics PGR110 Spring 2026

This assignment is worth 100% of the grade of the course and is graded A-F.

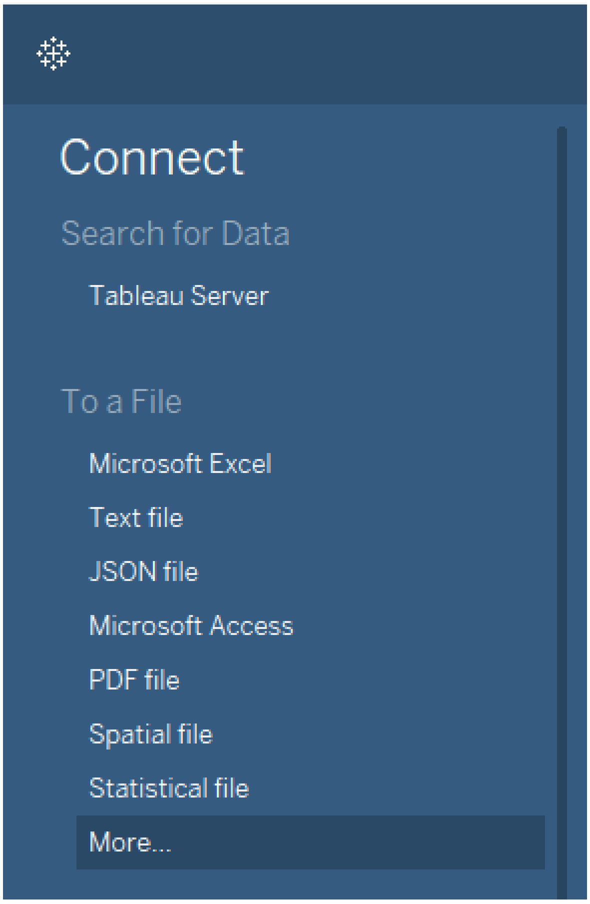
Group Exam (05.06-19.06)

The data:

Data is attached on WiseFlow.

File: World Development Indicators 2001-2025.xls

This file can be opened in Tableau Public Desktop Edition, either by dragging it into Tableau (and it will open automatically). Or through “Microsoft Excel”



The case:

You have just joined **a Nordic NGO/Non-Profit organization** as a **data analyst**. The organization has collected a dataset based on World Bank indicators for the five Nordic countries with key indicators such as internet usage, literacy, education, life expectancy, governance, GDP, and CO_2 emissions – to mention a few.

About the Organization

Nordic Digital Development Observatory (NDDO)

"We provide data-driven insights to help governments and international partners understand long-term trends in digital access, education, and sustainable development. Our goal is to support better-informed decisions by highlighting patterns, disparities, and opportunities across countries."

Your task is to **create an interactive dashboard in Tableau** that helps your colleagues, partners, and stakeholders:

- **Analyze**: Compare how countries perform across economic, social, and digital development.
- **Gain insight**: Identify patterns and relationships between the available measures. You are expected to select a subset of relevant indicators (e.g., 4–8 variables) that best support your analysis of digital, economic, and social development).
- **Interpret**: Provide evidence-based interpretations of patterns and relationships in the data and discuss what these may imply for digital development priorities.

Think of your dashboard as a **decision-support tool**: it should not only present the numbers but also tell a story that is clear, interactive, and useful for evidence-based discussion.

You should connect to the data, build different types of visualisations/charts, gather insight and combine those charts into an interactive dashboard (either exploratory or explanatory).

There is data over time, there is geographic data and there are different measures to analyze.

PS: if Tableau treats a Measure as a Dimension (text), change it to a number (decimal) and convert it to a Measure. Also remember to use correct aggregation (SUM is not always the answer).

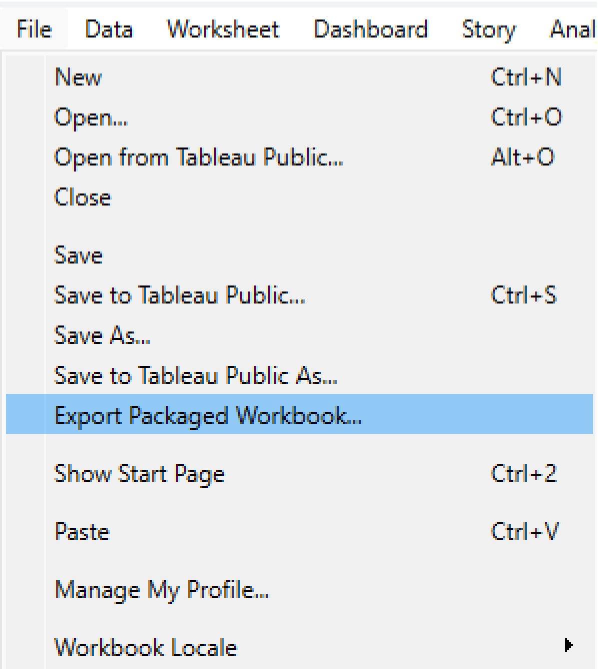
👉 Some examples to consider based on the data:

- Time series (lines/areas)
- Comparisons (bars/heatmaps)
- Relationships (scatter plots/bubbles)
- Geography (maps)
- Storytelling (dashboards with filters)

- **Possible analytical perspectives:**
    - How does digital access (e.g., broadband, mobile subscriptions) relate to economic development (GDP per capita)?
    - Are there observable relationships between education levels and digital infrastructure?
    - How do environmental indicators (CO_2 emissions) relate to economic growth?
    - Are Nordic countries converging or diverging over time?

Submission:

The finished **workbook** with the dashboard/charts must be shared like this:

1. Submit the Tableau workbook as a packaged workbook, .twbx, named GroupNumber.twbx.
    a. 

2. Submit the written report as a PDF.

The **formal report** should consist of two parts:

Part A) Report to the board committee of your NGO, where you should identify and communicate key trends, differences, and relationships in the data that are relevant for understanding digital and socio-economic development in the Nordic countries.

Discuss possible implications of these findings, while acknowledging limitations in the data and analysis.

You should support your discussion with examples from your dashboard (e.g., screenshots or annotated figures).

Part B) Here you should focus on answering what were the design principles and practices that you used to design your dashboard and how did you apply them.

This is done by using references from the course textbooks and other authoritative sources to explain the design principles and practices that you used to design your dashboard. Here it is important to not just state the design principles and practices but explain how you applied them to design your dashboard.

Things to include in this part of the report are

- A description of the dataset and data preparation in your own words. How did you prepare the data to analyze it in Tableau? What issues did you notice in the data? How did you address them?
- The dataset has been pre-filtered to include Nordic countries, but still requires preparation and validation and you should justify any decisions to exclude variables or time periods.
- A description of your final design choices using the 5 layers of data visualisation framework of Andy Kirk (Composition, Color, Annotation, Interactivity, Data Representation).
- A discussion of the evolution of your design from your first idea to your final design, using sketching to showcase your ideas and design iterations.
- Reflection on your dashboard development process. How was your dashboard design process? What challenges did you face and how did you handle them?

The report should be 6-10 pages in length (excluding title page and references but including data visualisations and other figures). It should be submitted as a PDF.

References

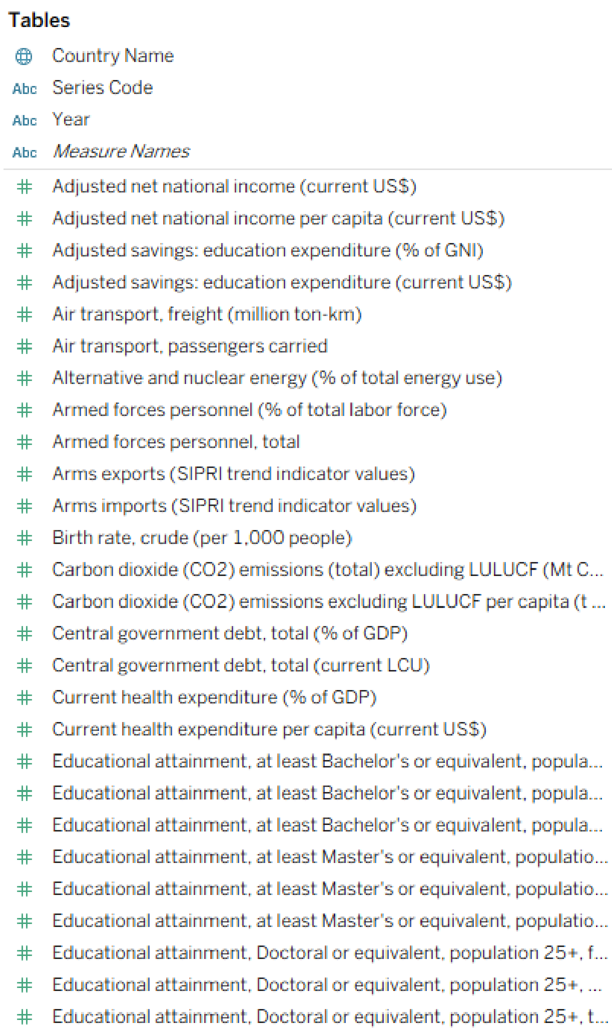
Your report references should be done in Vancouver style https://i.ntnu.no/en/academic-writing/vancouver, which uses numbered in-text citation. Each reference in the bibliography should be cited at least once in the report text. The course textbooks should be cited with references including specific chapters and page numbers.

Dataset

The dataset is from World Bank Group (World Development Indicators - https://www.worldbank.org/ext/en/home) and shows several indicators (measures) from the 5 Nordic countries (Denmark, Finland, Iceland, Norway, and Sweden) in the period 2001 to 2025. There might be some gaps/nulls in the data, important that this is considered and filtered away, measures missing several years' worth of data might be omitted completely.

There are 2 sheets in the Excel, one with the data used for the analysis (“WDI Data”) and one with the metadata -> explanations for all the indicators being used (“Metadata (not used for exam)”).

When the data is connected in Tableau, it should look something like this:



Guiding Evaluation Questions

The rubric “pgr110_rubric.docx” gives criteria for achieving grades A-F. These questions are a supplement to the rubric, and are designed to help you complete the exam.

Part 1) Interactive Dashboard

- Does the dashboard address the case?
- Is the dashboard easy to use?
- Can the user get the information they want?
- Is the design visually appealing?
- Is the Tableau file in a .twbx format (including data)?
- Does the dashboard:
    - Support comparison across countries and over time?

    - Enable user interaction (e.g., filters, highlighting)?
    - Clearly communicate key insights through visual design and layout?
    - Avoid unnecessary complexity or clutter?
- Does the dashboard include
    - At least 3 different chart types (it’s also possible to build several dashboards)?
    - time-based visualisation?
    - comparison across countries?
    - interactive features such as filter, parameter, highlight, or dashboard action?
    - A clear title and short explanatory text/annotations.
- Note that there is no single correct solution. The quality of your work will be evaluated based on clarity of insight, appropriateness of methods, and quality of design.

Part 2) Report to the Board Committee

- Does this part of the report address the case?
- Are the report’s analysis and interpretations supported by appropriate data visualisations?
- Is the visualized information easy to understand?
- Is there a clear point to the data visualisations, are they telling a story?

Part 3) What were the design principles and practices that you used to design your dashboard, and how did you apply them?

- Does this part of the report address the case?
- Does the report include all 4 description items (data description and preparation, 5 layers model, design process, and reflection)?
- Have relevant data visualisation design principles and theories been appropriately applied and referenced?
- Have the course textbooks been appropriately cited?